

3D picking for any industrial robot. A depth camera on the wrist and a small computer beside the robot find every part of the size you enter lying on a flat surface, number them in the pick order you choose, and bring the tool to the grip on each one. A connector puts this into the robot's own programs; the robot's program opens and closes the gripper.

Planning figures, not guarantees. The numbers below are deliberately conservative estimates derived from the software's limits, its programmed speeds and the camera's noise. This release is a beta and the figures have not been measured on a production cell. **Test with your parts, gripper and layout before you commit to them.**

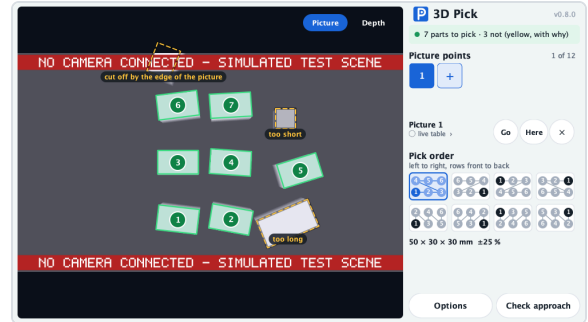
COMPATIBLE ROBOTS

Arm	6-axis articulated	Collaborative or industrial. Comes straight down; turns the tool to align the fingers.
Reach	$\geq 0.5 \text{ m}^E$	Rated. Every grip is checked against the arm's kinematics first.
Payload	+ 0.1 kg	Camera and bracket, centre of gravity ~40 mm off the flange axis.
Tool flange	ISO 9409-1-50-4-M6 ISO 9409-1-80-6-M8	Brackets for both, under the gripper. Others need an adapter.
Controller	Ethernet, TCP	Its program opens a TCP socket and moves to a computed pose.
Gripper	Two-finger	Any parallel gripper the robot's program opens and closes.

PICKING PERFORMANCE (by camera-to-part distance; verify on your cell)

	0.30 to 0.60 m working zone	1.0 to 1.5 m far limit	
Parts found	99 %^S	4 of 4^M	Right-size parts in view. Wrong ones picked: 0.
Size resolved to	$\pm 5 \text{ mm}$	$\pm 17 \text{ mm}$	It checks a size; it does not measure one.
Sizes told apart	$\geq 10 \text{ mm}^E$	$\geq 35 \text{ mm}^E$	Closer sizes read as the same part.
Smallest side	20 mm^E	45 mm^E	Edges blur by ~1 % of the distance.
Lowest part	5 mm^E	15 mm^M	Above the surface; rough ones raise it.
Part centre	$\pm 3 \text{ mm}^S$	$\pm 10 \text{ mm}^E$	95 %, plus calibration: 9 mm per 1°.
Part heading	$\pm 2.5^\circ^S$	$\pm 8^\circ^E$	95 %. None for a square top.
Gap to neighbours	$\geq 25 \text{ mm}$	$\geq 40 \text{ mm}^E$	20 mm finger room a side.
Time to find	0.9 s^M	1.1 s^M	Per picture, Raspberry Pi 5.
Distance limits	0.28 to 1.6 m	No depth nearer; nothing farther is picked.	
Grip position	$\pm 10 \text{ mm}^E$	Plan the jaws for it. Mostly calibration.	
Part pose	Any face down	Boxes on any face; cylinders on end.	
Cycle, node only	15 s per part^E	To the grip. ~2 parts/min with a short place.	

^M Measured with the camera computer and a D435, one setup. ^S Simulated: 720 parts through a model of the D435's noise. ^E Estimate, not a measured or guaranteed value.



The Universal Robots connector on a PolyScope 5 pendant (simulated scene).

CONNECTORS

Universal Robots PolyScope 5.4 to 5.26, URCap 0.9.0
PolyScope X 10.8 to 10.14, URCap 0.7.0

Other makes On request

WHAT A CELL NEEDS

Camera	RealSense D435 (848 × 480, 30 fps, USB 3) on a printed bracket
Computer	Raspberry Pi class, beside the robot; ships set up
Network	Ethernet; TCP 7621 and 7622
Setup	One camera calibration; optional pick areas
Software	Free, open source (MIT). Kit from Advin.

NOT FOR (LIMITS OF THE DEPTH CAMERA)

- Measuring parts or quality checks
- Defect, surface or print inspection
- Telling apart parts of nearly the same size
- Thin or flat parts: sheets, washers, labels
- Transparent, mirror-like or very dark parts
- Bin picking: piled or leaning parts
- Shapes other than boxes and upright cylinders
- Direct sunlight on the work area
- Driving the gripper or the place move

CAPABILITIES

- **Finds parts by volume, from depth alone:** boxes on any face, cylinders on end; the floor is fitted round each part.
- **Any position, any rotation.** Fingers across the short side, or the long one.
- **Eight pick orders:** by rows or by columns, from any corner.
- **Checks before it moves:** reach by the arm's kinematics, finger room.
- **Says why not:** near misses drawn in yellow with the reason.
- **Queues what it saw:** no trip back to the picture point.
- **Runs inside the robot's own program** through its connector. No internet.

Download and details: perceptronics.advin.io · **Kit and questions:** nick@advin.io · **Source:** github.com/JimothyJohn/perceptronics

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